

📌 What is Component Scanning in Spring?

- ◆ Component Scanning is a feature in Spring that automatically detects and registers Spring-managed beans in the Spring IoC container.
- ◆ Instead of creating objects manually or defining every bean in configuration, Spring scans packages and finds classes annotated with:

- 👉 @Component
- 👉 @Service
- 👉 @Repository
- 👉 @Controller
- 👉 @RestController

- ◆ Spring automatically creates objects (beans) for these classes and manages them.

Without Component Scanning:

```
public class EmployeeService {  
  
    public void getEmployee() {  
        System.out.println("Employee Details");  
    }  
}
```

- ◆ You would need to create the object manually:

```
EmployeeService service = new EmployeeService();  
service.getEmployee();
```

With Component Scanning:

- ◆ Step 1: Create a Component

```
package com.vijaystack.service;  
  
import org.springframework.stereotype.Component;  
  
@Component  
public class EmployeeService {
```

```

    public void getEmployee() {
        System.out.println("Employee Details");
    }
}

```

- ♦ Step 2: Enable Component Scanning

```

package com.vijaystack;

import org.springframework.context.annotation.ComponentScan;
import org.springframework.context.annotation.Configuration;

@Configuration
@ComponentScan("com.vijaystack")
public class AppConfig {
}

```

- ♦ Spring scans the `com.vijaystack` package and its sub-packages.
- ♦ Step 3: Use the Bean

```

package com.vijaystack;

import org.springframework.context.ApplicationContext;
import org.springframework.context.annotation.AnnotationConfigApplicationContext;

import com.vijaystack.service.EmployeeService;

public class Test {

    public static void main(String[] args) {

        ApplicationContext context =
            new AnnotationConfigApplicationContext(AppConfig.class);

        EmployeeService service =
            context.getBean(EmployeeService.class);

        service.getEmployee();
    }
}

```

Output:

Employee Details